DREW

Lecture 4: Data Representation

CSCI 260 Computer Systems & Architecture
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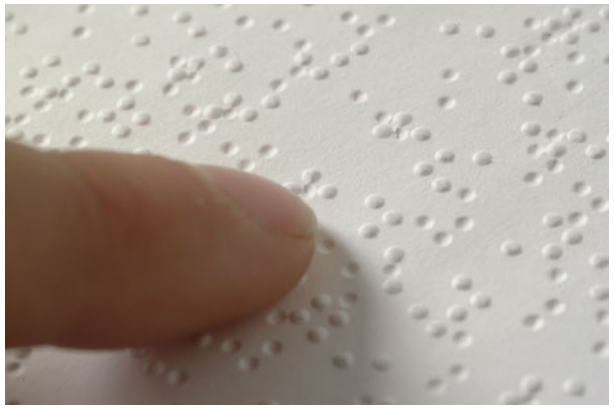
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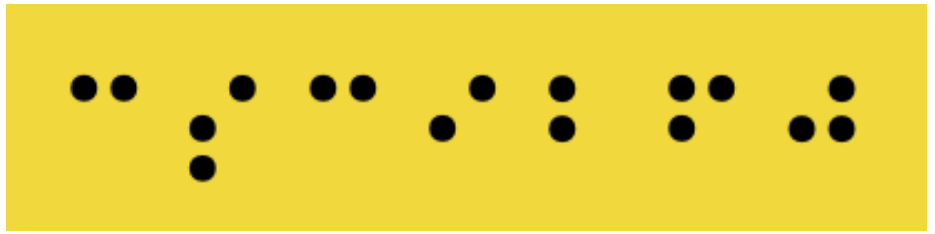
From Binary to Text

- **Known**: Everything in a digital computer is based on binary.
- **Question**: How are text represented in computers then?

- **Answer**: Binary, of course. (How??)

- A real-world example: The *Braille writing system* for visually impaired individuals so that they can read by touching the text.





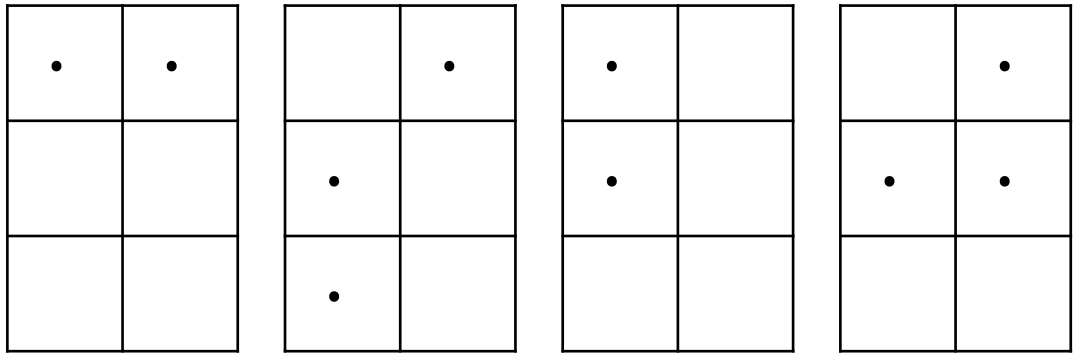
C S C I 2 6 0

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Number with Fractions: Floating Point

The *Braille writing system*

- Is based on a 3x2 grid, where a cell can either have a dot or not have a dot.
- Different dot combinations result in different letter/number (text) readings.



- Sounds somewhat familiar?

- Essentially, a 6-bit binary system:
  - dot in cell = 1 no dot in cell = 0
- Each combination is an *encoding* of the corresponding character.

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ASCII Code

- Text in digital computers are also encoded using binary.
- The *American Standard Code for Information Interchange* (*ASCII* code)
  - Is a 7-bit binary code: $2^7 = 128$ characters (*8-bit)
  - Contains *printable* characters: letters, numbers, special chars.;
    - "A" = 1000001 "i" = 1101001 "7" = 0110111 "{" = 1111011
  - and *non-printable* characters: ESC, labels: End of Text, etc.
- Provides a standard for hardware and software (OS, app); when a "4" is typed on keyboard, the CPU, monitor, etc., will know and react to that "4".

Hexadecimal Representation

- It is convenient to write binary strings as *hexadecimal* (base 16) strings.
  - 4 binary bits → 1 hexadecimal digit: less errors for us and for the machine.

Binary	Decimal	Hex
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7

Binary	Decimal	Hex
1000	8	8
1001	9	9
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F

Hexadecimal Representation

- 4 binary bits → 1 hexadecimal digit

- **Ex.** (Conversion): 01110101000111101001101011 1

0x 3 A 8 F 4 D 7

- Note that, this is not a new machine representation, just a convenient way to write the strings. Therefore, same operations.
  - Carry upon an $(F)_{16}$ (i.e., $(16)_{10}$).

- Many applications: ASCII, representing colors, instructions in ISA, etc.

ASCII Code

- ASCII is a 7-bit (*8-bit) binary code:
  - Max value: 0x7F or 01111111.
  - Extended full 8-bit ver. with 128 spec. char.

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E.2 Standard ASCII codes

Table E.2 The Standard ASCII Table

ASCII			ASCII			ASCII			ASCII		
Character	Dec	Hex	Character	Dec	Hex	Character	Dec	Hex	Character	Dec	Hex
nul	0	00	sp	32	20	@	64	40	'	96	60
soh	1	01	!	33	21	A	65	41	a	97	61
stx	2	02	"	34	22	B	66	42	b	98	62
etx	3	03	#	35	23	C	67	43	c	99	63
eot	4	04	$	36	24	D	68	44	d	100	64
enq	5	05	%	37	25	E	69	45	e	101	65
ack	6	06	&	38	26	F	70	46	f	102	66
bel	7	07	'	39	27	G	71	47	g	103	67
bs	8	08	(	40	28	H	72	48	h	104	68
ht	9	09	)	41	29	I	73	49	i	105	69
lf	10	0A	*	42	2A	J	74	4A	j	106	6A
vt	11	0B	+	43	2B	K	75	4B	k	107	6B
ff	12	0C	'	44	2C	L	76	4C	l	108	6C
cr	13	0D	-	45	2D	M	77	4D	m	109	6D
so	14	0E	.	46	2E	N	78	4E	n	110	6E
si	15	0F	/	47	2F	O	79	4F	o	111	6F
dle	16	10	0	48	30	P	80	50	p	112	70
dc1	17	11	1	49	31	Q	81	51	q	113	71
dc2	18	12	2	50	32	R	82	52	r	114	72
dc3	19	13	3	51	33	S	83	53	s	115	73
dc4	20	14	4	52	34	T	84	54	t	116	74
nak	21	15	5	53	35	U	85	55	u	117	75
syn	22	16	6	54	36	V	86	56	v	118	76
etb	23	17	7	55	37	W	87	57	w	119	77
can	24	18	8	56	38	X	88	58	x	120	78
em	25	19	9	57	39	Y	89	59	y	121	79
sub	26	1A	:	58	3A	Z	90	5A	z	122	7A
esc	27	1B	;	59	3B	[	91	5B	{	123	7B
fs	28	1C	<	60	3C	\	92	5C	|	124	7C
gs	29	1D	=	61	3D	]	93	5D	}	125	7D
rs	30	1E	>	62	3E	^	94	5E	~	126	7E
us	31	1F	?	63	3F	_	95	5F	del	127	7F

ASCII Code

- How large a file is?
  - Same as asking, how many characters are there in that file.

- The *width* of an ASCII character, 8-bit, is then called "1-Byte":
  the size a computer can "*bite*"/process at a time.
- 1000 chars.: ~1 kB,
- MB, GB, TB, etc.

Note: In networking, the unit is *bit*, NOT Byte.

1000Mb/s download speed = 125MB/s file write rate

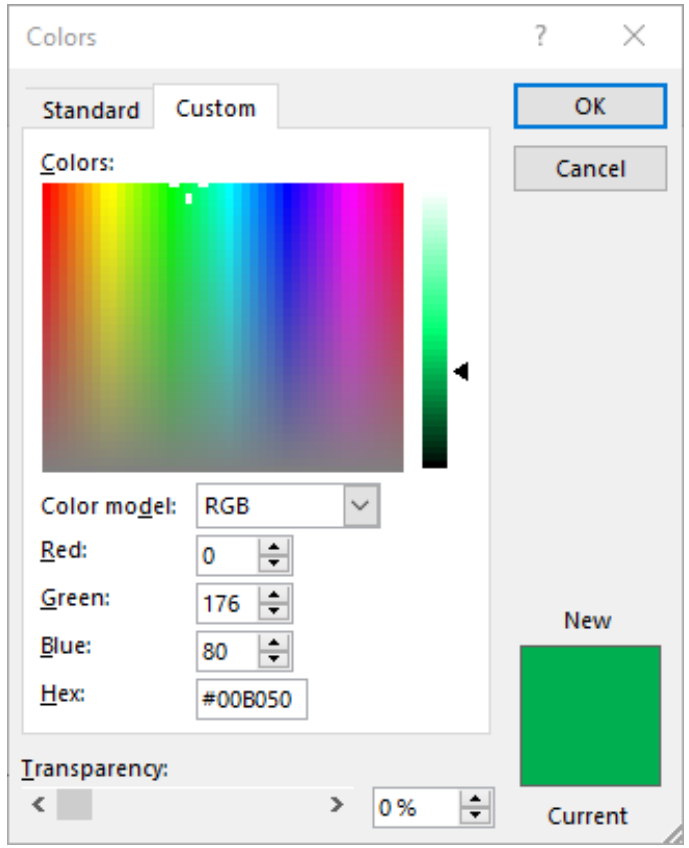
Other Codes

- However, computers help connect the world!
  - 128 characters are not enough: å, ê, ü, (Latin-based),
    α, π (Greek),
    字 (~3000 most used Chinese),
    ∞, ∫ (Mathematical),
    
- **Unicode** (UTF-8): let's have more bits!
  - Up to 21-bit, depending on the character.
  - Different char.s will have different byte sizes.

Other Data Types

- **More Binary Codes:** barcodes, QR codes, and Morse code

- ASCII for encoding **char.s** → **Strings**: sequence of char.s
- **Images**: arrays of pixels
  - B/W: 0/1
  - RGB: 8-bit per color
  - Etc.
- **Sound**: time-sequence of floating-point numbers for pitch, etc.
- **Videos**: images + sound



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Software for this course

- LC-3 (Little Computer-3) Simulator for Assembly Language
  - ~~LC-3 distributed with textbook~~
  - https://wchargin.com/lc3web/

  - It simulates on how *Instruction* saved/loaded at a memory location is executed.
  - By combining many instructions, desired computational goal is reached. (Remember that high/low-lan. achieve the same goal?)

  - **Ex.**: How to multiply two numbers with no multiplication instruction?
    There's ADD and can check whether something is zero instructions.

Software for this course

*: Not on Apple Silicons though!

- Linux Command Line Tools
  - Typically a Virtual Machine (VM) (you will for sure in the future anyways)
    - Virtual Box (Windows & Mac*)
    - VMware Fusion (Mac*), VMware Player (Windows)
  - with a Linux Operating System (OS)
    - Ubuntu
    - Kali
    - Red Hat, CentOS, & etc.
  - For us, CoCalc

Software for this course

- C Programming Language
  - **CLion** by JetBrains: free for education, register with your Drew email.
  - I need you to go ahead and install **CLion** (https://www.jetbrains.com/clion/), so I can determine if additional library is needed.
    - **Mac**: If already have Xcode, no need to worry at all.
      Otherwise, install Xcode via App Store.
    - **Windows**: May need to install **gcc** via **Cygwin**.

Questions?

- **Next Week**:
  - In-Class Exercises on Binary – Attendance
  - §3 Digital Logic Structure